

Converting ER Model to Relational Tables

Q. How is an ER Diagram mapped into a Relational Database schema Explain the rules for converting Entities, Attributes, and the various Relationships (1:1, 1:N, M:N) into database tables.

[DEF] Definition to Remember

ER-to-Relational Mapping is the process of converting a conceptual ER diagram into a physical Relational Model consisting of Tables, Columns, and Keys. Each entity, attribute, and relationship must be transformed using a specific set of mapping rules.

1. Rule 1: Strong Entities and Attributes

- * Every Strong Entity Type -> becomes a separate Table.

2. Rule 2: Multi-Valued Attributes

- * Cannot store a multi-valued attribute in the main table (relational tables require atomic values).

3. Rule 3: Weak Entities

- * Weak Entity -> converted into a separate table with its own attributes.

4. Rule 4: Mapping Binary Relationships

Cardinality	New Table Needed	Action
1:1	No	Add PK of one table as FK in the other
1:N	No	Add PK of the "One" side as FK in the
M:N	**Yes**	Create a Junction Table with both PKs

1:1 Mapping:

MANAGER (Mgr_ID) DEPARTMENT (Dept_ID, Mgr_ID as FK)

1:N Mapping:

DEPARTMENT (Dept_ID) EMPLOYEE (Emp_ID, Dept_ID as FK)

M:N Mapping:

STUDENT (Roll_No) ENROLLS (Roll_No FK, Course_ID FK) COURSE (Course_ID)

[MUST-WRITE] Must-Write Points (for 10 marks)

1. ER-to-Relational Mapping converts ER blueprints into physical Tables, Columns, and Keys.
2. Every Strong Entity -> one Table; its Key Attribute -> Primary Key.
3. Multi-Valued Attributes cannot exist in the main table -- a new table is created.
4. Weak Entity -> new table; PK = Owner Entity's PK (FK) + Partial Key.
5. 1:1 relationship -> no new table; place one PK as FK in the other.
6. 1:N relationship -> no new table; place "One" side's PK as FK in "Many" side.
7. M:N relationship -> requires a Junction (Associative) Table with both PKs as FKs.

[RECALL] Quick Recall

Strong Entity->Table (PK) -> Multi-valued->New Table -> Weak Entity->Table (FK+Partial Key) -> 1:1 (FK) -> 1:N (FK on Many) -> M:N (Junction Table)